

College of Industrial Technology
King Mongkut's University of Technology North Bangkok

เลขที่นั่งสอบ

Final Examination of Semester 2

Subject: 392152 Chemistry II

Date: 6 March 2014

Year: 2013

Section: 15-16

Time: 10.00-12.00

Name: _____ ID: _____ Class: _____

Instructions:

1. Cheating will result in failure of all classes registered for the current semester. Students who are caught cheating will also be denied registering for the following semester.
2. No documents are allowed to be taken out of the examination room.
3. Only calculators are allowed in the examination.
4. This exam is a closed book examination.
5. Electronic communication devices are **NOT** allowed in the examination room.
6. The examination has 7 pages (including this page), 2 sections and a total score of 66 points.
7. Write all your solution and answer on this examination sheet.

Solvent	Boiling Point ($^{\circ}\text{C}$)	K_b ($^{\circ}\text{C}/\text{m}$)	Freezing Point ($^{\circ}\text{C}$)	K_f ($^{\circ}\text{C}/\text{m}$)
Water	100	0.512	0	1.86
Benzene	80.1	2.53	5.5	5.12
Acetic acid	118.1	3.07	16.6	3.90
Nitrobenzene	210.9	5.24	5.7	7.00
Phenol	182	3.56	43	7.40
Camphor	207.4	5.61	178.4	40.0

Part 1. Multiple Choice (13 points)

Choose the letter of the choice in the answer table that would be the best completes the statement or answers the question.

Question no.	a	b	c	d
1				
2				
3				
4				
5				
6				

Question no.	a	b	c	D
7				
8				
9				
10				
11				
12				
13				

- What is electronic structure of Li? (shell1 shell2 shell3)
 a. 1 1 0 b. 2 1 0 c. 1 1 1 d. 2 0 1
- The name of group VIIA is
 a. noble gases b. alkaline earth metals
 c. inner transitions d. halogen
- The d-block elements are
 a. transition elements b. main group elements
 c. inner transition elements d. alkaline metal elements
- What is the most probable charge of the alkaline earth metal ions?
 a. +2 b. +4 c. -1 d. +3
- Oxides of most nonmetals combine with water to form _____.
 a. an acid b. a base c. water and a salt d. water
- What is not a colligative property?
 a. vapor pressure lowering b. boiling point depression
 c. freezing point depression d. osmotic pressure
- What volume of 16 M sulfuric acid must be used to prepare 1.6 L of a 0.10 M H_2SO_4 solution?
 a. 1 mL b. 10 mL c. 100mL d. 1000 mL
- Hydrogen and nitrogen react to form ammonia according to the reaction, $\text{H}_2 + \text{N}_2 \rightarrow \text{NH}_3$.
 If 3.0 moles of H_2 with 2.0 mol of N_2 are reacted, what is the limiting reagent?
 a. H_2 b. N_2 c. NH_3 d. More than one

9. In problem 8, how many moles of what reactant is left over?

- a. 2 mol of H_2 b. 1 mol of N_2 c. 2 mol of NH_3 d. 1 mol of H_2

10. Calculate the freezing point of the 1.25 m sucrose solution.

- a. $-2^\circ C$ b. $-1^\circ C$ c. $1^\circ C$ d. $2^\circ C$

11. If a theoretical yield for the reaction was 156 g and we actually made 122 g of the product, what is the percent yield?

- a. 78.2 % b. 128 % c. 19 % d. none of these

12. If we dilute 5 mL of 0.15 M NaCl to final volume of 5 L, what is the final concentration of NaCl?

- a. 15.00 M b. 0.15 M c. 0.015 M d. 0.00015 M

13. A hydrocarbon sample has an empirical formula of CH and the molar mass of this compound is 78 g/mol. What is the chemical formula of this compound?

- a. CH_4 b. C_5H_{10} c. C_2H_2 d. C_6H_6

Part 2 Answer (53 points)

1. Give two examples of elements for each category. (2 points)

- a) noble gases _____ b) halogens _____
c) alkali metals _____ d) alkaline earth metals _____

2. Write the oxide and chloride forms and acid-base behavior for the following elements (7 points)

	Na	Mg	Al	Si	P	S	Cl
Oxide							
Oxide acid-base behavior							
Chloride							
Chloride acid-base behavior							

3. Please write the mathematical formula for the dilution of solution and explain the concept shortly. (3 points)

4. What volume of 8 M hydrochloric acid must be used to prepare 3.0 L of a 0.50 M hydrochloric acid solution? (2 points)

5. A 50.0 g sample of NaOH is dissolved in 0.600 L of water. What volume of this solution would be needed to create a 1.5 L solution that is 0.200 M NaOH?(Na=23, O=16, H=1) (3 points)

6. A stock solution containing $C_6H_{12}O_6$ was prepared by dissolving 36 g of $C_6H_{12}O_6$ in water to a final volume of 1.000 L. The solutions A were then prepared by diluting 50.00 mL of stock solution to 1000.0 mL. Calculate the concentrations of the stock solution and solutions A. (C=12, H=1, O=16) (5 points)

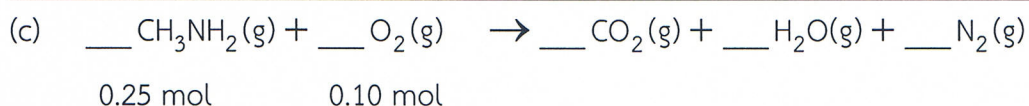
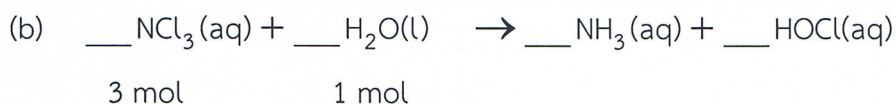
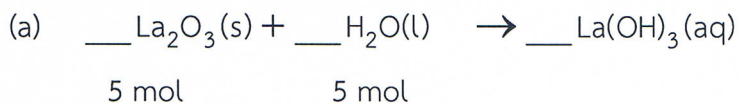
8. When 15.0 grams of ethyl alcohol, C_2H_5OH , is dissolved in 750. grams of formic acid, the freezing point of the solution is $7.20^\circ C$. The freezing point of pure formic acid is $8.40^\circ C$. Evaluate K_f for formic acid. (C=12, H=1, O=16) (4 points)

e solution is 7.20°C . The freezing point of
ic acid. ($C=12$, $H=1$)

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[illegible]

12. Balance the following equations by providing the missing coefficients. If the amount of each reactant used in the laboratory is given below, identify the limiting reagent in each case. (6 points)



13. In the reaction of chlorine (Cl_2) and sodium (Na) to form sodium chloride (NaCl), if 30.0 g of chlorine react with 30.0 g of sodium determine (Na=23, Cl=35.5) (5 points)
- the mass of sodium chloride produced
 - the amount of excess reactant left

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